Math 2270 Review for §2.3 – 2.5

Ex • Find elimination matrices that put A in to an upper triangular form U

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 1 \\ -2 & 2 & 0 \end{bmatrix} \qquad E_1 \cdot E_2 \cdot E_3 A = U$$

• Factor $A = L \cdot U$ where L is lower triangular

Ex • Write the 3×3 permutation matrices for

1) exchange first 2 rows

2) exchange rows 1 & 3

3) send 1st row to 2nd row, 2nd " to 3rd row 3rd " " 1st row.

• Find inverses

• Do 1–3 for columns (what side do we multiply by?)

Ex • Find the inverse of $A = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & 1 & t \end{pmatrix}$

• Solve $A\vec{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

Ex Prove that a) $(AB)^{-1} = B^{-1}A^{-1}$, b) If B_1, B_2 are inverses of A then $B_1 = B_2$, c) $(A+B)^2 = A^2 + 2AB + B^2$ iff $AB = BA$. d) If A & B are triangular (upper) then so is AB.

Ex Show that $\begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix}$ is invertible if $a \neq 0$ & $a \neq b$